A Diagrammatic Proof of the Haldane Gap via Spacetime Tensor Networks

(Dated: October 8, 2026)

Following the recent machine-assisted resolution of the Haldane conjecture for periodic and boundary-field spin-1 Heisenberg chains by OpenAI [1, 2], we show that the analytical derivations can be recast as elementary tensor-network contractions. Formulating the system as an exact continuous-time thermal matrix product operator exposes the underlying spacetime geometry. The step from a finite-scale estimate to a uniform gap then reduces to a contractive scale-doubling recurrence on planar diagrams, under three pages long and formalized in Lean 4 with the two numerical inputs as its only unproved hypotheses. The finite-scale estimate itself is taken unchanged from Ref. [1], whose certificate uses spectral filters and character projections. This illustrates how diagrammatic representations can clarify and compress machine-generated proofs in quantum many-body physics.

We consider the antiferromagnetic spin-1 Heisenberg chain with periodic boundary conditions $\boldsymbol{S}_{L+1} = \boldsymbol{S}_1$ and $J = k_B = 1$:

$$H_L = \sum_{j=1}^{L} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}, \qquad Z_L(\beta) = \operatorname{Tr} e^{-\beta H_L}. \tag{1}$$

We take even $L \geq 4$ so the transfer trace is an even power. Let $E_0(L) \leq E_1(L) \leq \cdots$ include multiplicities, with gap $\Delta_L = E_1(L) - E_0(L)$ ( ) and Gibbs weights $w_m = e^{-\beta E_m(L)}/Z_L(\beta)$:

$$\frac{w_1}{w_0} = e^{-\beta\Delta_L} \leq e^{-c\beta} \quad \Longrightarrow \quad \Delta_L \geq c. \tag{2}$$

The ratio decays in β like ; the task is to make its rate c independent of L:

Theorem 1 (Periodic gap [1]). *For every even $L \geq 2304$, the chain (1) has a unique ground state and $\Delta_L > (\log 20)/784 \simeq 0.003821$.*

Theorem 1 was established by OpenAI [1] via a thirty-page coordinate construction using degree-12 spectral filters and dihedral character decompositions. After showing how a thermal purity near one forces a gap, our proof has three stages: (i) an exact continuous-time thermal MPO yields a self-adjoint spatial transfer operator; (ii) spacetime contraction identities couple spatial and thermal purities so joint scale doubling quadratically suppresses both defects (the *spacetime bootstrap*); and (iii) interpolating between successive doubled lengths covers all even $L \geq 2304$.

Why tensor-network diagrams? In Eq. (1), L counts sites in a sum whereas β sits in an exponential. Drawing $Z_L(\beta)$ as a two-dimensional tensor network [3, 4] puts space and imaginary time on equal footing on a torus: vertical double legs ‖ carry the physical spin space through imaginary time (trace Tr; blue, thermal ρ), horizontal legs — couple neighboring sites in the virtual bond space (trace $\mathrm{Tr_v}$; red, spatial σ), joined legs are contracted, and closed loops are traces. Reading the same network along either axis turns the proof into planar diagram identities; Appendix A caps the horizontal ends with boundary vectors for an open chain.

From purity to gap.— For a positive trace-one operator with eigenvalues $p_0 \geq p_1 \geq \cdots$, the purity obeys

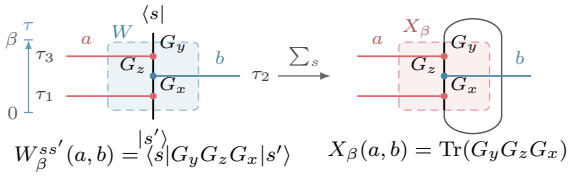
$$P = \sum_i p_i^2 \leq p_0 \sum_i p_i = p_0. \tag{3}$$

Thus the *purity defect* $1 - P$ bounds $1 - p_0 \leq 1 - P$: a purity near one forces a spectrum like with a single dominant bar. For the Gibbs state, $P > 1/2$ implies a unique ground state, with $1 - w_0 \leq 1 - P$ and $e^{-\beta\Delta_L} = w_1/w_0 \leq (1-P)/P$. To reach Eq. (2) it suffices to show that $1 - P$ falls like $e^{-c\beta}$ uniformly in L. Squaring in imaginary time improves purity at fixed L [Eq. (10)], but growing L adds thermal excitations; advancing L and β together requires a spatial transfer operator.

Thermal MPO.— As a warm-up, consider the classical Ising ring with spins $t_j = \pm 1$. Its Boltzmann weight $\prod_j e^{\beta t_j t_{j+1}}$ is a product of entries of the 2×2 matrix $K_{tt'} = e^{\beta t t'}$, and each spin appears in exactly two neighboring factors. Summing a spin shared by two factors is a matrix product, $\sum_{t_2} K_{t_1 t_2} K_{t_2 t_3} = (K^2)_{t_1 t_3}$, which is what a joined leg means; summing every spin multiplies K around the ring, and the closure $t_{n+1} = t_1$ turns the last sum into a trace, $Z_n = \operatorname{Tr} K^n$. Here we construct a real symmetric operator X_β on the virtual bond space that plays the role of K, with $Z_n(\beta) = \mathrm{Tr_v}\, X_\beta^n$. In the Cartesian basis $s, s' \in \{x, y, z\}$, $(S^\alpha)_{ss'} = -i\epsilon_{\alpha ss'}$ gives real operators $G_\alpha = iS^\alpha$ with entries $(G_\alpha)_{ss'} = \epsilon_{\alpha ss'}$: $G_\alpha v = v \times e_\alpha$ turns the plane perpendicular to e_α by a quarter turn and annihilates e_α (for example $G_z e_x = -e_y$ and $G_z^2 = \operatorname{diag}(-1, -1, 0)$), so $\|G_\alpha\| = 1$. Because $G_\alpha \otimes G_\alpha = i^2 S^\alpha \otimes S^\alpha$, $-H_L = \sum_{j,\alpha} G_{\alpha,j} G_{\alpha,j+1}$. Since the simplex $0 < \tau_1 < \cdots < \tau_k < \beta$ in imaginary time τ has volume $\beta^k/k!$, $e^{\beta A} = \sum_{k \geq 0} \int_{0<\tau_1<\cdots<\tau_k<\beta} d\tau_1 \cdots d\tau_k\, A^k$. For $A = -H_L$, each factor selects a bond and a label α, inserting G_α on both sites of that bond at time τ. We call such a factor an *event* (τ, α) on its bond, and the finite set of events on one bond its *bond history*; each term of the expansion is a choice of history on every bond. For the histories a and b of the left and right bonds of a site, the local MPO tensor multiplies the matrices hitting its worldline in time order ($\mathcal{T}_\tau$ placing later times to the left):

$$a \;\; \overset{s}{\underset{s'}{W}} \;\; b = W_\beta^{ss'}(a,b) = \left\langle s \middle| \mathcal{T}_\tau \prod_{(\tau,\alpha)\in a\cup b} G_\alpha \middle| s' \right\rangle. \tag{4}$$

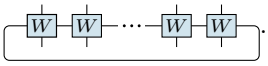
For $a = \{(\tau_1, x), (\tau_3, y)\}$ and $b = \{(\tau_2, z)\}$ with $\tau_1 < \tau_2 < \tau_3$, the site worldline and its physical trace $\mathrm{Tr}_s = \mathrm{Tr}$ are



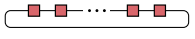
Operators on different sites commute, so once the events on every bond are fixed, the time-ordered product over the ring splits into one worldline per site, whose physical trace is $X_\beta(a,b) = \sum_s W_\beta^{ss}(a,b)$. The bond history plays the role of the Ising spin, and X_β the role of K. Summing the three labels $\alpha \in \{x, y, z\}$ and integrating ordered times defines a measure μ_β of mass $\sum_{k\geq 0}(3\beta)^k/k! = e^{3\beta}$ (unit mass on the empty history). With a_j the history of bond $(j, j+1)$ and $a_0 = a_n$,

$$Z_n(\beta) = \int \prod_{j=1}^{n} d\mu_\beta(a_j) \prod_{j=1}^{n} X_\beta(a_{j-1}, a_j), \tag{5}$$

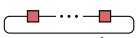
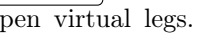
and integrating each shared history a_j is the matrix product of the Ising ring, now on $L^2(\mu_\beta)$. Contracting n tensors around the ring recovers the convergent exponential series as an exact MPO on $L^2(\mu_\beta)$ without bond truncation [3, 4]:

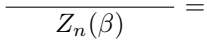
$e^{-\beta H_n} =$  . (6)

The left leg a is the row index of X_β and the right leg b its column index, so symmetry means that a site cannot tell its left bond from its right one. Because a and b enter Eq. (4) through $a \cup b$ and each event inserts the same real matrix G_α at both ends of its bond (*bond reflection positivity*, $-\boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1} = \sum_\alpha G_{\alpha,j} G_{\alpha,j+1}$), the spatial transfer kernel X_β is real and symmetric ($X_\beta(a,b) = X_\beta(b,a)$). The condition is not automatic: for a bond term $-h = Q \otimes P$ with $Q = \mathrm{diag}(1,1,0)$ and $P = \mathrm{diag}(1,0,0)$, a site receives P through its left bond and Q through its right one, so a history a of two events, with the empty history $\emptyset$ on the other bond, gives $X_\beta(a, \emptyset) = \mathrm{Tr}\, P^2 = 1$ from the left but $X_\beta(\emptyset, a) = \mathrm{Tr}\, Q^2 = 2$ from the right. In general, regrouping the four legs of $-h$ by site (rows $s_j s'_j$, columns $s_{j+1} s'_{j+1}$) turns $-h = \sum_\alpha G_\alpha \otimes G_\alpha$ into the 9×9 matrix $\sum_\alpha \mathrm{vec}\, G_\alpha\, \mathrm{vec}\, G_\alpha^{\mathsf{T}}$, a sum of squares; a positive semidefinite regrouped bond is the reflection positivity that the general criterion requires [5]. Since $\|G_\alpha\| = 1$ on $\mathbb{C}^3$, $|X_\beta(a,b)| \leq \mathrm{Tr}\, I = 3$; with the finite mass $e^{6\beta}$ of $\mu_\beta \times \mu_\beta$, X_β is self-adjoint and Hilbert–Schmidt on $L^2(\mu_\beta)$, giving a discrete real spectrum $|\lambda_0(\beta)| \geq |\lambda_1(\beta)| \geq \cdots$ ($\sum_i \lambda_i(\beta)^2 < \infty$). Closing the ring of transfer vertices gives, for even n,

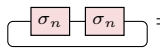
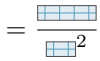
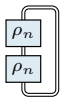
$Z_n(\beta) =$  $= \mathrm{Tr_v}\, X_\beta^n = \sum_i |\lambda_i(\beta)|^n,$ (7)

so raising the eigenvalues to the nth power concentrates the weight on λ_0:  ($n = 4$).

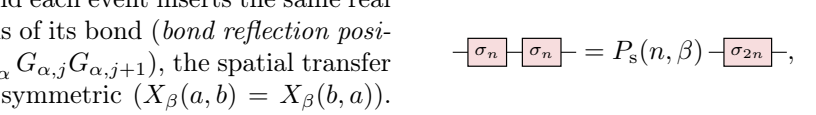
Spatial and thermal purities.— The transfer operator gives space its own density operator. Cut the ring of Eq. (7) at one bond,  ↦  : what is left is X_β^n with two open virtual legs. Cutting the torus instead along one time slice leaves $e^{-\beta H_n}$ with open physical legs, Eq. (6). Normalizing both to unit trace gives

$$\sigma_n(\beta) = \frac{\text{  }}{Z_n(\beta)} = \frac{X_\beta^n}{Z_n(\beta)}, \qquad \rho_n(\beta) = \frac{e^{-\beta H_n}}{Z_n(\beta)}. \tag{8}$$

For even n, both operators are positive with unit trace; denote their eigenvalues by $p_i = |\lambda_i(\beta)|^n / Z_n(\beta)$ and w_m, respectively. Their purities join two copies, along space for σ_n and along imaginary time for ρ_n. Drawing $Z_n(\beta)$ as a spacetime torus  of width n and height β (dashed edges identified), both are one ratio of tori:

$$\begin{aligned} P_{\mathrm{s}}(n,\beta) &= \mathrm{Tr_v}\, \sigma_n^2 = \text{  } = \frac{\mathrm{Tr_v}\, X_\beta^{2n}}{Z_n(\beta)^2} \\ &= \frac{Z_{2n}(\beta)}{Z_n(\beta)^2} = \frac{\text{  }}{\text{  }^2}, \\ P_\tau(n,\beta) &= \mathrm{Tr}\, \rho_n^2 = \text{  } = \frac{\mathrm{Tr}\, e^{-2\beta H_n}}{Z_n(\beta)^2} \\ &= \frac{Z_n(2\beta)}{Z_n(\beta)^2} = \frac{\text{  }}{\text{  }^2}. \end{aligned} \tag{9}$$

Squaring.— Squaring improves a purity: squaring σ_n doubles the length; squaring ρ_n doubles the inverse temperature. Indeed $\sigma_n^2 = X_\beta^{2n}/Z_n^2 = (Z_{2n}/Z_n^2)\, \sigma_{2n}$, and likewise for ρ_n^2:



(10)

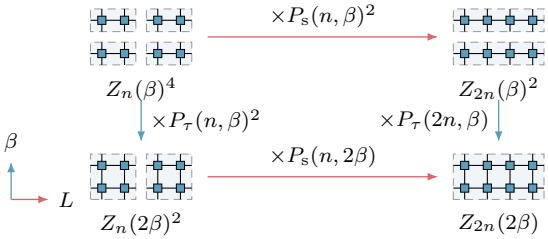
Both replace the normalized eigenvalues p_i by p_i^2/P, concentrating the spectrum:  ↦ . Since $1 - P = 2\sum_{i<k} p_i p_k$, dropping cross terms from $(\sum_{i<k} p_i p_k)^2$ bounds the new defect $1 - P'$ via $f(z) := z^2/[2(1-z)^2]$:

$$1 - P' = \frac{2\sum_{i<k} p_i^2 p_k^2}{P^2} \leq \frac{(1-P)^2}{2P^2} = f(1-P). \tag{11}$$

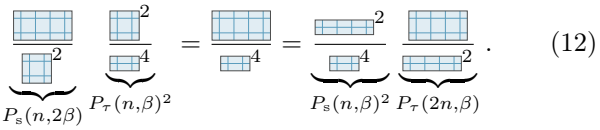
Because f increases on $[0,1)$ with $f(z) \sim z^2/2$, squaring roughly squares the purity defect $1 - P$, keeping it far below the diagonal,  for $0 \leq z \leq .3$.

Spacetime coupling identities.— Each squaring doubles one side of the torus: space ($n \to 2n$),  ↦  , or imaginary time ($\beta \to 2\beta$),  ↦  . Because a uniform gap needs both sides to grow together, we need identities trading purity in one direction for purity in the other. Each gluing multiplies by exactly one purity, because each purity is a ratio of tori [Eq. (9)]: $Z_{2n}(\beta) = P_{\mathrm{s}}(n,\beta)\, Z_n(\beta)^2$ and $Z_n(2\beta) = P_\tau(n,\beta)\, Z_n(\beta)^2$. Starting from four tori $Z_n(\beta)^4$, gluing pairs first in space and then in time, or first in time and then in space, reaches the

same doubled torus $Z_{2n}(2\beta)$ (red arrows glue in space, blue in time, labeled by the purity factor they multiply by):



Writing each purity as its ratio of tori, Eq. (9), the two paths cancel to the same quotient:



(12)

Solving for one factor, and repeating at length $2n$, gives

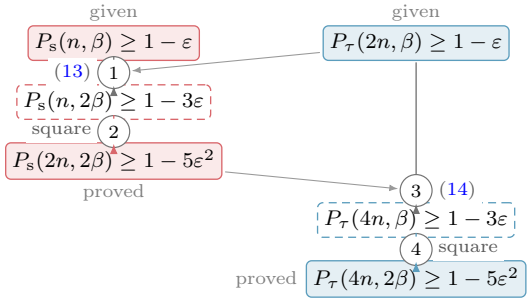
$$P_s(n,2\beta) = \frac{P_s(n,\beta)^2 P_\tau(2n,\beta)}{P_\tau(n,\beta)^2}, \tag{13}$$

$$P_\tau(4n,\beta) = \frac{P_\tau(2n,\beta)^2 P_s(2n,2\beta)}{P_s(2n,\beta)^2}. \tag{14}$$

Doubling estimate.— Alternating the two coupling identities with the two squarings doubles the torus in both directions ( ) and roughly squares the purity defect. For even n and $0 \le \varepsilon \le 1/100$,

$$\begin{aligned} & P_s(n,\beta),\ P_\tau(2n,\beta) \ge 1-\varepsilon \\ \Longrightarrow\quad & P_s(2n,2\beta),\ P_\tau(4n,2\beta) \ge 1-5\varepsilon^2. \end{aligned} \tag{15}$$

Four moves prove Eq. (15). Read the diagram below from top to bottom (red boxes bound P_s, blue P_τ; solid given or proved, dashed intermediate; circles label the identity and grey arrows supply the second input):



The proved pair is the given pair at $(2n,2\beta)$, so Eq. (15) iterates.

Move 1 (coupling). Since $P_\tau(n,\beta) \le 1$, Eq. (13) gives $P_s(n,2\beta) \ge (1-\varepsilon)^3 \ge 1-3\varepsilon$. *Move 2* (squaring in space). For even n, $\sigma_n(2\beta)$ has nonnegative spectrum and squares to $\sigma_{2n}(2\beta)$. Since $1-(1-\varepsilon)^3 \le 3\varepsilon$ and $9/[2(1-3\varepsilon)^2] \le 5$ for $\varepsilon \le 1/100$, Eq. (11) and monotonicity of f give $P_s(2n,2\beta) \ge 1-\delta$, with

$$\delta = f\big(1-(1-\varepsilon)^3\big) = \tfrac{1}{2}\big[(1-\varepsilon)^{-3}-1\big]^2 \le 5\varepsilon^2 \le \varepsilon. \tag{16}$$

Move 3 (coupling). Equation (14) with $P_s(2n,\beta) \le 1$ and $\delta \le \varepsilon$ gives $P_\tau(4n,\beta) \ge (1-\varepsilon)^2(1-\delta) \ge (1-\varepsilon)^3$. *Move 4* (squaring in time). Squaring $\rho_{4n}(\beta)$ to $\rho_{4n}(2\beta)$ gives $P_\tau(4n,2\beta) \ge 1-\delta \ge 1-5\varepsilon^2$, proving Eq. (15).

Starting estimate and iteration.— The doubling estimate needs one starting scale where both defects are below $1/100$. Proposition 5.5 of Ref. [1] gives the $j=0$ starting estimates:

$$P_s(2304,784),\ P_\tau(4608,784) \ge 0.99. \tag{17}$$

Equation (17) cannot come from diagonalizing X_β, which acts on whole bond histories; Ref. [1] uses only traces of small rings and one variational bound. It shifts $H_L \to H_L + \kappa LI$, with $\kappa = 700741/500000$ chosen so that $-\kappa$ lies just above the energy per site of a matrix-product trial state ψ on $N=72$ sites; the factor $e^{-\kappa\beta L}$ in $Z_L(\beta)$ cancels from both purity ratios (9). Since $Z_N(\beta)$ is a sum of positive Boltzmann factors and the variational principle gives $E_0(N) \le \langle\psi|H_N|\psi\rangle/\langle\psi|\psi\rangle < -\kappa N$, the ground-state factor alone gives $Z_N(\beta) > 1$. Combined with eigenvalue bounds from the twisted traces $\mathrm{Tr}_v(X_\beta^n U_g)$ of rings of at most twelve sites, with a spin rotation U_g of the histories on one virtual bond, this forces a dominant λ_0; Appendices I–M of the companion certificate note draw these networks and six doublings from $(36, 49/4)$ to Eq. (17). After $j \ge 0$ doublings, set

$$n_j = 2304\,2^j, \qquad \beta_j = 784\,2^j, \qquad \varepsilon_j = \frac{20^{-2^j}}{5}. \tag{18}$$

Since $\varepsilon_0 = 1/100$ and $\varepsilon_{j+1} = 5\varepsilon_j^2$, Eq. (15) gives $P_s(n_j,\beta_j), P_\tau(2n_j,\beta_j) \ge 1-\varepsilon_j$ by induction on j: each inductive step runs Moves 1–4 once. The defect falls doubly exponentially in j, $\varepsilon_0 = 10^{-2} \to \varepsilon_4 \approx 3\times 10^{-22}$ (log scale), and so exponentially in β_j, $\varepsilon_j = \frac{1}{5}\exp[-\beta_j(\log 20)/784]$, as needed in Eq. (2).

All even lengths.— These estimates hold at the doubled lengths n_j and $2n_j$. To cover every even $L \ge 2304$, choose j with $n_j \le L \le 2n_j$. Since the weights $p_i = |\lambda_i(\beta)|^n/Z_n(\beta) \in [0,1]$ sum to one, any power $r \ge 1$ gives $Z_{rn}(\beta)/Z_n(\beta)^r = \sum_i p_i^r \le \sum_i p_i = 1$; taking $r = L/n_j \ge 1$ at β_j and $r = 2n_j/L \ge 1$ at $2\beta_j$ yields

$$Z_L(\beta_j) \le Z_{n_j}(\beta_j)^{L/n_j}, \qquad Z_L(2\beta_j) \ge Z_{2n_j}(2\beta_j)^{L/(2n_j)}. \tag{19}$$

Dividing $Z_L(2\beta_j)$ by $Z_L(\beta_j)^2$, substituting the torus identity $Z_{2n_j}(2\beta_j)/Z_{n_j}(\beta_j)^4 = P_\tau(2n_j,\beta_j)P_s(n_j,\beta_j)^2$ [Eq. (12)], and using $L/(2n_j) \le 1$ with $(1-\varepsilon_j)^3 \ge 1-3\varepsilon_j > \frac{1}{2}$ gives

$$P_\tau(L,\beta_j) \ge \big[P_\tau(2n_j,\beta_j)P_s(n_j,\beta_j)^2\big]^{L/(2n_j)} \ge 1-3\varepsilon_j. \tag{20}$$

Recovering the gap.— With $w_0 \ge w_1 \ge \cdots$ for $\rho_L(\beta_j)$, Eqs. (3) and (20) give $w_0 \ge 1-3\varepsilon_j > 1/2$: the Gibbs spectrum looks like ▙ rather than ▙▄, so the ground state is unique and $w_1 \le 1-w_0 \le 3\varepsilon_j$. Since $3\varepsilon_j/(1-3\varepsilon_j) \le$

$(60/97)20^{-2^j} < 20^{-2^j}$,

$$e^{-\beta_j \Delta_L} = \frac{w_1}{w_0} \le \frac{1-w_0}{w_0} \le \frac{3\varepsilon_j}{1-3\varepsilon_j} < 20^{-2^j}. \tag{21}$$

Taking logarithms and dividing by $\beta_j = 784\,2^j$ gives, for every even $L \ge 2304$,

$$\boxed{\Delta_L > \frac{\log 20}{784} \simeq 0.003821.} \tag{22}$$

This proves Theorem 1, conditional on Eq. (17); in particular $\liminf_{L\to\infty,\, L\,\mathrm{even}} \Delta_L \ge (\log 20)/784 > 0$. The scale-doubling recurrence and the gap deduction are formally verified in Lean 4 (`lean/CertifiedGaps/PurityCriterion.lean`).

DISCUSSION: REPRESENTATION IN MACHINE-ASSISTED PROOFS

The uniform spectral gap theorems for the periodic and boundary-field spin-1 Heisenberg chains, together with the certified starting estimates, come from the machine-assisted proofs of OpenAI [1, 2]. Those thirty-page proofs locate eigenvalues and connect scales through explicit coordinate constructions, such as degree-12 polynomial spectral filters and dihedral character projectors, alongside lemmas unused in the final inequality. Automated provers work by local symbolic rewriting in a fixed basis; without the geometric pictures of many-body physics, their long chains of deductions naturally build up intermediate algebra to stay sound.

Here, the authors asked GPT-6 to re-examine the argument in terms of spacetime tensor networks and worked through the resulting structure interactively. Treating imaginary time and space alike on a torus reduces the self-adjoint spatial transfer operator and the contractive scale-doubling iteration to short identities between planar diagrams. Tensor-network diagrams are thus useful not only for calculation, but also for reading, simplifying, and extracting the physical mechanism of a machine-generated proof. The authors checked all mathematical and formal proofs.

[1] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Proposition 5.5 and Appendices A–C.

[2] OpenAI, *A boundary-field gap for the spin-one Heisenberg chain*, manuscript dated September 24, 2026, Secs. 2–6 and Appendices A and B.

[3] W. Tang, H.-H. Tu, and L. Wang, Continuous matrix product operator approach to finite temperature quantum states, Phys. Rev. Lett. **125**, 170604 (2020).

[4] B. Pirvu, V. Murg, J. I. Cirac, and F. Verstraete, Matrix product operator representations, New J. Phys. **12**, 025012 (2010).

[5] Companion note, *Scope of the spacetime bootstrap* (`thermal-mpo-gap-scope.tex`, 2026), Appendix E.

[6] T. Kennedy, Exact diagonalisations of open spin-1 chains, J. Phys.: Condens. Matter **2**, 5737 (1990).

Appendix A: Open chain with boundary fields

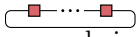
The networks above also bound the gap of the open chain with equal endpoint fields. On n spin-1 sites, with $\sum_\alpha (S^\alpha)^2 = 2I$, let

$$H_n^{\mathrm{s}} = \sum_{i=1}^{n-1} \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} - h(S_1^z + S_n^z), \qquad h = \frac{3}{5}. \tag{A1}$$

Theorem 2 (Boundary-field gap [2])**.** *For every odd $n \geq 1921$, the Hamiltonian (A1) has a unique ground state and $\Delta_n > (\log 10)/392 \simeq 0.005874$.*

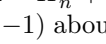
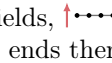
Boundary fields are essential here: without them, each end of the open spin-1 chain carries an effective spin-1/2 boundary mode, and the four lowest states are split only by an amount exponentially small in n [6]; equal fields lift this boundary degeneracy on odd chains [2]. As on the ring, by Eqs. (2) and (3), to prove Theorem 2 it suffices to bound the equal-field thermal purity $P_\tau^{\mathrm{s}}(n,\beta) = \sum_m w_m^2$ from below:

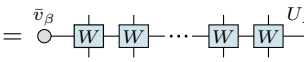
$$e^{-\beta\Delta_n} = \frac{w_1}{w_0} \leq \frac{1 - P_\tau^{\mathrm{s}}(n,\beta)}{P_\tau^{\mathrm{s}}(n,\beta)}. \tag{A2}$$

Strategy.— The ring proof cannot be applied directly because an open chain has no ring to close: its partition function is not a trace of X_β^n, so there is no spatial purity to square. Its bulk, however, shares the periodic transfer operator X_β, and on an open chain the outer virtual legs of the first and last sites are closed by boundary vectors (the quantum counterpart of $u^{\mathsf{T}} K^{n-1} u$ for an open classical Ising chain, with $u = (e^{\beta h}, e^{-\beta h})$ for a field h on each end spin). Whereas a periodic trace  weights every eigenvector of X_β by one, an open chain  weights the leading eigenvalue $\lambda(\beta)$ by the squared boundary overlap $c(\beta)$ [Eq. (A8)] and vanishes if the two are orthogonal. For long chains the open partition function is approximately $c(\beta)\lambda(\beta)^n$, so its thermal purity is approximately $[\lambda(2\beta)/\lambda(\beta)^2]^n c(2\beta)/c(\beta)^2$: a bulk factor controlled by the ring times an n-independent boundary factor that cancels when comparing lengths M and $2M$. Doubling the length therefore costs only the bulk factor, while squaring in imaginary time (10) needs no ring. Because this requires the leading term to dominate the remainder, the proof tracks the *boundary overlap* alongside the thermal purity through a boundary defect e that shrinks as the chain grows.

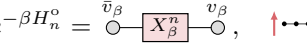
Boundary vectors.— The open chain uses the tensor W_β of Eq. (4), with the per-site energy shift $\kappa = 700741/500000$ of Ref. [2] (called a there) absorbed into each tensor as a factor $e^{-\kappa\beta}$, i.e., $H_n^\nu \to H_n^\nu + \kappa n I$; this makes the leading periodic eigenvalue $\lambda_0(\beta_j)$ close to one at the certified scales and cancels from every Gibbs state and purity ratio. On an end site, the field $\pm hS^z = \mp(ih)G_z$ enters the imaginary-time expansion as half a bond: for a history a with $|a|$ events, define the boundary vector on $L^2(\mu_\beta)$ by

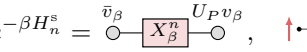
$$v_\beta(a) = \begin{cases} (ih)^{|a|}, & \text{all labels are } z, \\ 0, & \text{otherwise,} \end{cases} \qquad \|v_\beta\|^2 = e^{\beta h^2}, \tag{A3}$$

with value one on the empty history ($|a| = 0$); the z-event time integrals give $\|v_\beta\|^2$ as the sum of $(\beta h^2)^{|a|}/|a|!$ over $|a| \geq 0$. The conjugated left cap $\bar{v}_\beta$ weights each z event by $(-ih)G_z = +hS^z$, adding $-hS_1^z$ to the Hamiltonian, whereas the unbarred right cap v_β weights each z event by $(ih)G_z = -hS^z$ and adds $+hS_n^z$, giving the opposite-field chain $H_n^{\mathrm{o}} := H_n^{\mathrm{s}} + 2hS_n^z$,  . The π rotation $P = \mathrm{diag}(1,-1,-1)$ about x flips $S^z \to -S^z$ and every z label, so its virtual action U_P reverses the right field and restores equal fields,  . Opening the ring of Eq. (6) and capping its ends therefore gives

$$e^{-\beta H_n^{\mathrm{s}}} = \text{  }, \tag{A4}$$

and tracing the physical legs onto the transfer chain X_β^n of Eq. (8) gives the open-chain partition functions (the open counterparts of Z_n in Eq. (7)):

$$Y_n(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{o}}} = \text{  }, \tag{A5}$$

$$Y_n^{\mathrm{s}}(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{s}}} = \text{  }. \tag{A6}$$

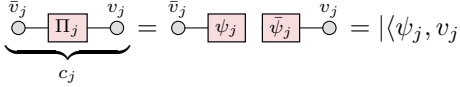
The opposite-field chain has identical bra and ket caps, so its even moments are nonnegative; this is why the proof works with H_n^{o} and transfers to H_n^{s} at the end. Simultaneous spin rotations of the histories act on $L^2(\mu_\beta)$ as real orthogonal operators commuting with X_β; in particular $[U_P, X_\beta] = 0$. Before the κ shift, the absolute series for the traced expansion sums to at most $3^n e^{\beta[3(n-1)+2h]} < \infty$, so Fubini's theorem justifies regrouping the time-ordered series into the site-by-site contractions (A5)–(A6).

Transfer operator spectrum.— The open-chain argument needs the following facts about X_β, all supplied by the ring. At successive inverse temperatures $\beta_{j+1} = 2\beta_j$, the ring estimates of Appendix B show that the leading eigenvalue $\lambda_j := \lambda_0(\beta_j)$ of Eq. (7) is simple and positive. Normalizing λ_j to one splits X_{β_j}/λ_j into its leading projector Π_j and orthogonal remainder R_j ($\Pi_j R_j = R_j \Pi_j = 0$):

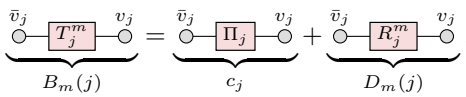
$$T_j := X_{\beta_j}/\lambda_j = \Pi_j + R_j, \qquad \Pi_j = |\psi_j\rangle\langle\psi_j|, \quad \|\psi_j\| = 1, \tag{A7}$$

so $T_j^m = \Pi_j + R_j^m$, with $\|R_j\| < 1$, $U_P \Pi_j = \Pi_j$, and $0 < r_j := \lambda_{j+1}/\lambda_j^2 \leq 1$ [Eqs. (B1)–(B2)].

Boundary overlap and residual decomposition.— We now split the open-chain contraction into the leading part, which carries the boundary overlap, and a remainder. Put $v_j = v_{\beta_j}$. Inserting the leading projector disconnects the network:

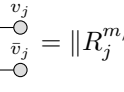
$$\text{  }, \tag{A8}$$

The overlap c_j is the squared projection of v_j onto ψ_j, ; unlike a periodic trace, where $\mathrm{Tr}_{\mathrm{v}}\,\Pi_j = 1$, it vanishes when the two are orthogonal. For even $m \geq 2$, $T_j^m = \Pi_j + R_j^m$ splits the normalized partition function $B_m(j) := Y_m(\beta_j)/\lambda_j^m > 0$ into c_j and a remainder $D_m(j)$:



(A9)

Since R_j is self-adjoint, $R_j^m = R_j^{m/2} R_j^{m/2}$ folds the chain at its midpoint into a squared norm, showing that the remainder is nonnegative:

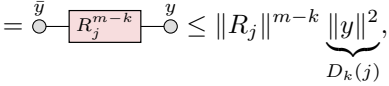
$$D_m(j) = \text{  } = \|R_j^{m/2} v_j\|^2 \geq 0. \tag{A10}$$

The boundary defect is defined by

$$e_m(j) := \frac{D_m(j)}{B_m(j)} = 1 - \frac{c_j}{B_m(j)}, \tag{A11}$$

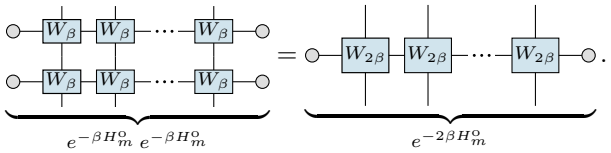
the grey fraction of the bar, , drawn at the starting value $e_{142}(2) \leq .14$ of Eq. (C1). Any $e_m < 1$ guarantees a nonzero overlap with the leading eigenvector.

Spatial remainder suppression.— A longer chain shrinks the remainder while leaving the leading overlap c_j unchanged. For even $m \geq k$, moving $k/2$ factors of R_j onto each boundary vector gives $y = R_j^{k/2} v_j$ and

$$D_m(j) = \bar{y} \text{  } y \leq \|R_j\|^{m-k} \underbrace{\|y\|^2}_{D_k(j)}, \tag{A12}$$

decaying geometrically by $\|R_j\|^{m-k}$ and ensuring $c_j \leq B_m(j) \leq B_k(j)$.

Thermal operator composition.— Because Eq. (A4) is an exact operator identity on the physical Hilbert space, composing $e^{-\beta H_m^{\mathrm{o}}} e^{-\beta H_m^{\mathrm{o}}} = e^{-2\beta H_m^{\mathrm{o}}}$ in imaginary time couples temperatures:



(A13)

This identity holds between the operators; it does not assert that two W_β tensors fuse into one $W_{2\beta}$ site by site. Normalizing the trace defines the opposite-field Gibbs state $\rho_m^{\mathrm{o}}(\beta) = e^{-\beta H_m^{\mathrm{o}}}/Y_m(\beta)$ and its thermal purity:

$$P_\tau^{\mathrm{o}}(m, \beta_j) := \frac{Y_m(\beta_{j+1})}{Y_m(\beta_j)^2} = \mathrm{Tr}\,\rho_m^{\mathrm{o}}(\beta_j)^2 \in (0, 1]. \tag{A14}$$

Substituting $Y_m(\beta_j) = \lambda_j^m B_m(j)$ and $\lambda_{j+1} = r_j \lambda_j^2$ into Eq. (A14) links adjacent temperatures:

$$B_m(j+1) = P_\tau^{\mathrm{o}}(m, \beta_j) r_j^{-m} B_m(j)^2. \tag{A15}$$

Coupled overlap recurrence.— This is Move 1: a small boundary defect at scale j gives a small one at scale $j+1$. Assuming $e_k(j) < 1$ at scale j, let primes denote $j+1$, choose even $m > k \geq 2$, and let $\|R_{j+1}\|^{m-k} \leq q < 1$. Spatial suppression at $j+1$ gives $D'_m \leq q D'_k$. Since c' is length-independent, $D'_k - D'_m = B'_k - B'_m$; substituting $D'_k = D'_m + B'_k - B'_m$ into $D'_m \leq q D'_k$ gives the first inequality, and dividing by B'_m the second:

$$(1-q) D'_m \leq q(B'_k - B'_m), \qquad e'_m \leq \frac{q}{1-q}\left(\frac{B'_k}{B'_m} - 1\right). \tag{A16}$$

Using $P_\tau^{\mathrm{o}}(k, \beta_j), r_j \leq 1$ and $B_m(j) \geq c_j = [1 - e_k(j)] B_k(j) > 0$, composition (A15) bounds the ratio at the previous temperature:

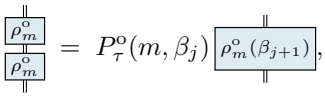
$$\frac{B'_k}{B'_m} = \frac{P_\tau^{\mathrm{o}}(k, \beta_j)}{P_\tau^{\mathrm{o}}(m, \beta_j)} \left[\frac{B_k(j)}{B_m(j)}\right]^2 r_j^{m-k} \leq \frac{1}{P_\tau^{\mathrm{o}}(m, \beta_j)[1 - e_k(j)]^2}. \tag{A17}$$

Because only the previous overlap appears in the denominator, $e'_m < 1$ establishes that the updated overlap remains positive. When $e_k(j)$ and $1 - P_\tau^{\mathrm{o}}(m, \beta_j)$ are $\mathcal{O}(x)$, choosing $m - k$ with $q \leq \mathcal{O}(x)$ ensures that e'_m contracts quadratically as $\mathcal{O}(x^2)$.

Thermal purity propagation.— This is Move 2: with the overlap under control, the length can double at almost no cost in purity. Since $B_{2M}(j+1) \geq c_{j+1} = [1 - e_M(j+1)] B_M(j+1)$ and $B_M(j) \geq B_{2M}(j)$ [Eq. (A12)], solving Eq. (A15) for $P_\tau^{\mathrm{o}}(m, \beta_j)$ and dividing the $m = 2M$ identity by the $m = M$ identity for even M yields

$$\frac{P_\tau^{\mathrm{o}}(2M, \beta_j)}{P_\tau^{\mathrm{o}}(M, \beta_j)} = r_j^M \underbrace{\frac{B_{2M}(j+1)}{B_M(j+1)}}_{\geq 1 - e_M(j+1)} \underbrace{\left[\frac{B_M(j)}{B_{2M}(j)}\right]^2}_{\geq 1}. \tag{A18}$$

Because $r_j^M \approx 1$ [Eq. (B2)], purity remains controlled upon doubling length. Squaring the Gibbs operator advances temperature, as in Eq. (10):



$$= P_\tau^{\mathrm{o}}(m, \beta_j) \tag{A19}$$

so the squaring bound (11) at $m = 2M$ drives $1 - P_\tau^{\mathrm{o}}$ to $\mathcal{O}(x^2)$.

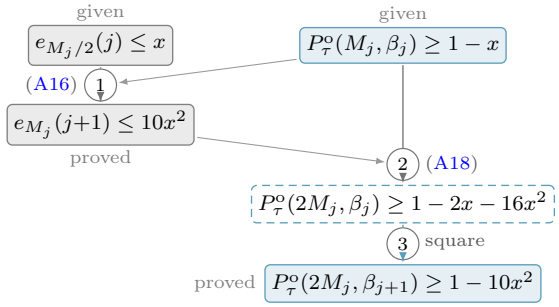
Scale induction.— Together, the overlap and purity bounds give an induction in which length and inverse temperature double at each step at the scales

$$\beta_j = 49\, 2^{j-2}, \quad n_j = 120\, 2^{j-2}, \quad M_j = 2n_j \quad (j \geq 2), \tag{A20}$$

with boundary-defect lengths $k_2 = 142$ and $k_{j+1} = M_j$. Using the ring purity defect ε_j ($\varepsilon_2 = .052$, $\varepsilon_{j+1} = 6\varepsilon_j^2$) bounded in Appendix B [Eq. (B1)] and the starting estimates at $j = 2$ in Appendix C, three explicit steps reach $j = 5$, and induction on $j \geq 5$ proves

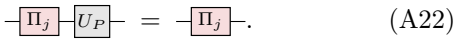
$$\varepsilon_j,\ e_{k_j}(j),\ 1 - P_\tau^{\mathrm{o}}(M_j, \beta_j) \leq x_j \quad (j \geq 5), \qquad x_5 = .01, \qquad x_{j+1} = 10 x_j^2, \qquad x_j = 10^{-1-2^{j-5}}. \tag{A21}$$

For $j \geq 3$, $k_j = M_j/2$, leaving $M_j/2$ sites to suppress remainders. One inductive step $j \to j+1$ is three moves. In the diagram below, read from top to bottom, each box is one bound: grey for the boundary defect $e_k(j)$, blue for the opposite-field thermal purity P^{o}_τ. Solid boxes are given at scale j or proved at scale $j+1$, the dashed box is intermediate, and the grey arrow entering a move is the second bound it uses:



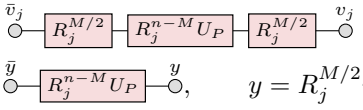
Move 1 is the overlap recurrence (A16)–(A17), move 2 the length doubling (A18), and move 3 the squaring (A19). Since $M_{j+1} = 2M_j$, the proved pair is the given pair of scale $j+1$, so the step repeats. The defect falls doubly exponentially, $x_5 = 10^{-2} \to x_9 = 10^{-17}$ (log scale); since $\beta_j = 392\,2^{j-5}$, it decays exponentially in inverse temperature.

Equal boundary fields and odd lengths.— The opposite-field estimates extend to the equal-field Hamiltonian (A1) via the spin-rotation symmetry U_P. Next to the leading projector, the rotation operator acts trivially:



(A22)

Thus $Q_j(n) := Y^{\mathrm{s}}_n(\beta_j)/\lambda_j^n$ shares the leading value c_j. For any even $M \leq n$, commuting U_P through the right-hand tensors ($[U_P, R_j] = 0$) isolates the remainder:



(A23)

Because $\|R_j^{n-M} U_P\| \leq 1$,

$$|Q_j(n) - c_j| \leq \|y\|^2 = D_M(j), \tag{A24}$$

bounding the odd-length remainder by $D_M(j)$ without requiring positivity of R_j^n.

For $M_j \leq n \leq 2M_j$ and $M = M_j = k_{j+1}$, Eq. (A24) gives $Q \leq c + D_M = B_M$ and $Q \geq c - D_M = (1 - 2e_M)B_M$; with $e_M(j+1) \leq x_{j+1}$ from Eq. (A21), this yields

$$Q_j(n) \leq B_M(j), \quad Q_{j+1}(n) \geq (1 - 2x_{j+1}) B_M(j+1) > 0. \tag{A25}$$

Since $P^{\mathrm{s}}_\tau(n, \beta_j) = Q_{j+1}(n)\lambda_{j+1}^n / [Q_j(n)\lambda_j^n]^2$, inserting these bounds and Eq. (A15) at $m = M$, and using $r_j^{n-M} \geq r_j^M$ [Eq. (B2)] with $\varepsilon_j \leq x_j$, $\varepsilon_{j+1} \leq 6x_j^2$, $x_{j+1} = 10x_j^2$, gives the equal-field purity:

$$\begin{aligned} P^{\mathrm{s}}_\tau(n, \beta_j) = \frac{Y^{\mathrm{s}}_n(\beta_{j+1})}{Y^{\mathrm{s}}_n(\beta_j)^2} &\geq (1 - 2x_{j+1}) P^{\mathrm{o}}_\tau(M, \beta_j) r_j^{n-M} \\ &\geq (1 - 2x_{j+1})(1 - x_j)(1 - \varepsilon_j)(1 - \varepsilon_{j+1}) \\ &\geq 1 - 2x_j - 26x_j^2 \geq 1 - 4x_j. \end{aligned} \tag{A26}$$

Proof of Theorem 2. The intervals $[M_j, 2M_j]$ cover every integer $n \geq M_5 = 1920$. For odd $n \geq 1921$, choose $j \geq 5$ with $M_j \leq n \leq 2M_j$. The largest Gibbs weight satisfies $w_0 \geq P^{\mathrm{s}}_\tau(n, \beta_j) \geq 1 - 4x_j > 1/2$: the spectrum looks like  , so the ground state is unique. Equation (A2) gives

$$e^{-\beta_j \Delta_n} \leq \frac{4x_j}{1 - 4x_j} \leq \frac{5}{12}\, 10^{-2^{j-5}} < 10^{-2^{j-5}}. \tag{A27}$$

Taking logarithms and using $\beta_j = 392\,2^{j-5}$ yields $\Delta_n > (\log 10)/392$. □

Appendix B: Transfer bounds at the open-chain scales

For the scales in Eq. (A20), Appendix A uses

$$\begin{aligned} \|R_j\|^{n_j} \leq d(\varepsilon_j), \qquad d(\varepsilon) = \frac{\varepsilon}{2 - 3\varepsilon}, \\ \varepsilon_2 = .052, \qquad \varepsilon_{j+1} = 6\varepsilon_j^2, \end{aligned} \tag{B1}$$

and

$$0 < r_j \leq 1, \quad r_j^{M_j} \geq (1 - \varepsilon_j)(1 - \varepsilon_{j+1}), \quad U_P \Pi_j = \Pi_j. \tag{B2}$$

These follow from the first row of Eq. (C1) by the doubling estimate of the main text, with a larger starting defect: if $P_{\mathrm{s}}(n, \beta), P_\tau(2n, \beta) \geq 1 - \varepsilon$, move 1 of Eq. (15) gives $P_{\mathrm{s}}(n, 2\beta) \geq (1 - \varepsilon)^3$, and squaring with Eq. (11) bounds $1 - P_{\mathrm{s}}(2n, 2\beta)$ by

$$f(1 - (1 - \varepsilon)^3) = \frac{\varepsilon^2}{2} \left[\sum_{s=1}^{3} (1 - \varepsilon)^{-s} \right]^2 \leq 6\varepsilon^2 \tag{B3}$$

for $0 \leq \varepsilon \leq .052$, in place of $5\varepsilon^2$. Since $6\varepsilon^2 \leq \varepsilon$, moves 3 and 4 give the same bound for $1 - P_\tau(4n, 2\beta)$. Induction therefore proves $P_{\mathrm{s}}(n_j, \beta_j), P_\tau(M_j, \beta_j) \geq 1 - \varepsilon_j$.

To turn the purity bound $P_{\mathrm{s}}(n_j, \beta_j) \geq 1 - \varepsilon_j$ into the remainder bound $\|R_j\|^{n_j} \leq d(\varepsilon_j)$ in Eq. (B1), consider a probability list with purity at least $1 - \varepsilon > 1/2$ and largest weight p_0; put $z = 1 - p_0$. The purity is at most $p_0^2 + z^2 = 1 - 2z(1 - z)$, so $2z(1 - z) \leq \varepsilon$. It is also at most p_0, so $p_0 > 1/2$: the largest weight is unique and $z < 1/2$. Then $z \leq 2z(1 - z) \leq \varepsilon$, hence $z \leq \varepsilon/[2(1 - \varepsilon)]$ and $z/(1 - z) \leq d(\varepsilon)$. Applied to the eigenvalues $p_i = |\lambda_i(\beta_j)|^{n_j}/Z_{n_j}(\beta_j)$ of $\sigma_{n_j}(\beta_j)$, the largest-modulus eigenvalue is simple and

$$\sum_{i \geq 1} \left| \frac{\lambda_i(\beta_j)}{\lambda_j} \right|^{n_j} \leq d(\varepsilon_j), \tag{B4}$$

which implies Eq. (B1). Positivity of $Z_N(\beta_j)$ for all large odd N ensures that the leading eigenvalue is positive, since a negative leading eigenvalue would make $\mathrm{Tr_v}\, X_{\beta_j}^N$ negative. The residual sum tends to zero since X_{β_j} is Hilbert–Schmidt and its residual spectral radius is strictly smaller than λ_j.

It remains to show $U_P\Pi_j = \Pi_j$ in Eq. (B2). The real kernel admits real orthogonal virtual actions of the three π rotations and the cyclic permutation of the coordinate axes, all commuting with X_{β_j}. On its real one-dimensional leading eigenspace these actions are signs. The order-three cyclic rotation has sign $+1$ and conjugates the three π rotations into one another, so their signs agree; since the product of two is the third, their common sign is $+1$, giving $U_P\Pi_j = \Pi_j$.

Finally, to bound $r_j = \lambda_{j+1}/\lambda_j^2$ in Eq. (B2), note that $Z_N(\beta_{j+1}) \le Z_N(\beta_j)^2$ since the thermal purity $P_\tau(N,\beta_j)$ is at most one. Taking Nth roots along even $N \to \infty$ gives $\lambda_{j+1} \le \lambda_j^2$. For the lower bound at the required finite length, the largest eigenvalue of $\sigma_{M_j}(\beta_{j+1})$ is at least $1-\varepsilon_{j+1}$, since $M_j = n_{j+1}$. Hence

$$\begin{aligned}\lambda_{j+1}^{M_j} &\ge (1-\varepsilon_{j+1}) Z_{M_j}(\beta_{j+1}) \\ &\ge (1-\varepsilon_{j+1})(1-\varepsilon_j) Z_{M_j}(\beta_j)^2 \\ &\ge (1-\varepsilon_{j+1})(1-\varepsilon_j)\lambda_j^{2M_j},\end{aligned}$$

which is the remaining bound in Eq. (B2).

Appendix C: Starting estimates for the open chain

The induction of Appendix A [Eq. (A21)] starts at $j = 5$, where all three defects are below $x_5 = .01$, whereas the certified inputs are given at $j = 2$. This appendix bridges $j = 2$ to $j = 5$ in four steps: it states the inputs, turns the three moves of Appendix A into numerical update rules, runs those rules for three steps, and verifies the induction step for $j \ge 5$.

Inputs.— The certified starting estimates combine periodic variational estimates with small-cluster boundary certificates from Ref. [2]:

$$\begin{aligned} P_{\mathrm{s}}(120,49) \ge .948, \qquad & P_\tau(240,49) \ge .948, \\ e_{142}(2) \le .14, \qquad & P_\tau^{\mathrm{o}}(240,\beta_2) \ge .79. \end{aligned} \tag{C1}$$

The first row gives the ring purity defect $\varepsilon_2 = .052$ at $(n_2,\beta_2) = (120,49)$ used in Appendix B; the second row gives the open-chain boundary defect at $k_2 = 142$ and opposite-field thermal purity at $M_2 = 240$.

Update rules.— Write E_j for an upper bound on $e_{k_j}(j)$ and η_j for an upper bound on $1 - P_\tau^{\mathrm{o}}(M_j,\beta_j)$. Move 1 controls the overlap: combining Eqs. (A16) and (A17) gives

$$\begin{aligned} e_m(j+1) &\le F(q, e_k(j), P_\tau^{\mathrm{o}}(m,\beta_j)), \\ F(q,e,t) &= \frac{q}{1-q}\left[\frac{1}{(1-e)^2 t} - 1\right], \end{aligned} \tag{C2}$$

which increases with q, e and decreases with t. The suppression factor q must bound $\|R_{j+1}\|^{M_j - k_j}$; since Eq. (B1) bounds $\|R_{j+1}\|^{n_{j+1}}$ and $n_{j+1} = M_j$, this fixes the exponent $(M_j - k_j)/M_j$ in Eq. (C3). Moves 2 and 3 control the purity via Eqs. (A18) and (11). With $k_2 = 142$ and $k_{j+1} = M_j$, the updates are

$$q_j = d(\varepsilon_{j+1})^{(M_j - k_j)/M_j}, \tag{C3}$$

$$E_{j+1} \ge F(q_j, E_j, 1-\eta_j), \tag{C4}$$

$$\eta_{j+1} \ge f\big(1 - (1-\eta_j)(1-\varepsilon_j)(1-\varepsilon_{j+1})(1-E_{j+1})\big), \tag{C5}$$

for any $E_{j+1}, \eta_{j+1} \in [0,1)$ satisfying these inequalities.

Three explicit steps.— Starting from $\varepsilon_2 = .052$, $E_2 = .14$, and $\eta_2 = 1 - .79 = .21$, three applications of Eqs. (C4)–(C5) give the rational upper bounds:

j	E_j	η_j	q_j (upper bound)
2	.14	.21	.142
3	.12	.155	.0282
4	.016	.026	.0028
5	.001	.001	—

Both defects collapse along the iteration, E_j:  and η_j:  ($j = 2,\dots,5$, each relative to $j = 2$). Rational arithmetic confirms these step bounds (e.g., at $j = 2$ the exponent in Eq. (C3) is $(240-142)/240 = 49/120$, so $q_2 \le .142$ amounts to $d(\varepsilon_3)^{49} < .142^{120}$, with subsequent steps verified by squaring), and $\varepsilon_5 < .000015$, establishing the base case $j = 5$ of Eq. (A21).

The induction step.— Assuming Eq. (A21) at scale $j \ge 5$ with $x = x_j \le .01$, the relation $k_j = M_j/2$ gives

$$\begin{aligned} q_j &\le \sqrt{\frac{6x^2}{2 - 18x^2}} \le 2x, \\ F(2x, x, 1-x) &\le \frac{6x^2}{(1-2x)(1-x)^3} < 7x^2. \end{aligned} \tag{C6}$$

Setting $E_{j+1} = 10x^2$ satisfies Eq. (C4). The four-factor product in Eq. (C5) is at least $1 - 2x - 16x^2$, and $f(2x + 16x^2) < 3x^2$, so $\eta_{j+1} = 10x^2$ satisfies Eq. (C5) with $\varepsilon_{j+1} \le 6x^2$, completing the induction for Eq. (A21).